

Lab: Cassandra CRUD Operations



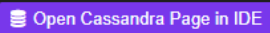
Objectives

After completing this lab you will be able to:

- Insert data into a table with an INSERT command
- Read data from a table by querying from it
- Update and delete data from the table based on specific criteria

Exercise 1 - Getting the environment ready

Open the Cassandra database page by clicking the button below:



On that page, click the **Create** button to create a Cassandra database.

After creation has finished, click the **Cassandra CLI** button (below the **Create** button from the previous step) to connect to the database using the cqlsh CLI.

1. Problem:

Create a keyspace named training using SimpleStrategy and replication factor of 3.

▼ Click here for Hint

use CREATE KEYSPACE command with appropriate options

▼ Click here for Solution

On the cqlsh run the below command.

```
1 CREATE KEYSPACE training
2 WITH replication = {'class': 'SimpleStrategy', 'replication_factor' : 3};
```

2. Problem:

Create a table named movies with columns 'movie_id' which is the primary key and integer type, 'movie_name' of type text, and 'year_of_release' of type integer.

▼ Click here for Hint

use CREATE TABLE command with appropriate options

▼ Click here for Solution

On the cqlsh run the below command.

```
1 use training;
2 CREATE TABLE movies(
3 movie_id int PRIMARY KEY,
4 movie_name text,
5 year_of_release int
6 );
```

Exercise 2 - Insert data into a table

Let us insert a row into the table movies.

On cqlsh run the below command.

```
1 INSERT into movies(
2 movie_id, movie_name, year_of_release)
3 VALUES (1, 'Toy Story', 1995);
```

Verify that the data is saved.

```
1 select * from movies;
```

Insert the below movies into the table.

movie_id	movie_name	year_of_release
2	Jumanji	1995
3	Heat	1995
4	Scream	1995
5	Fargo	1996

▼ Click here for Solution

On the cqlsh run the below command.

```
1  INSERT into movies(  
2  movie_id, movie_name, year_of_release)  
3  VALUES (2,'Jumanji',1995);  
4  
5  INSERT into movies(  
6  movie_id, movie_name, year_of_release)  
7  VALUES (3,'Heat',1995);  
8  
9  INSERT into movies(  
10 movie_id, movie_name, year_of_release)  
11 VALUES (4,'Scream',1995);  
12  
13 INSERT into movies(  
14 movie_id, movie_name, year_of_release)  
15 VALUES (5,'Fargo',1996);
```

Exercise 3 - Read data from a table

In the previous exercise you have inserted some data into the table named **movies**.
Let us query the data in the **movies** table.

Query all rows.

```
1  select * from movies;
```

Query the movie name with movie_id 1.

```
1  select movie_name from movies where movie_id = 1;
```

Exercise 4 - Update data in a table

The movie_id for Scream is 4. It was released in 1996 and not 1995.
Here is how you modify it.

```
1 UPDATE movies
2 SET year_of_release = 1996
3 WHERE movie_id = 4;
```

Verify that the update was successful.

```
1 select * from movies where movie_id = 4;
```

Exercise 5 - Delete data from a table

Delete the movie with the movie_id 5.

```
1 DELETE from movies
2 WHERE movie_id = 5;
```

Verify using the below command.

```
1 select * from movies;
```

Practice exercises

1. Problem:

Insert the below movie into the movies table.

movie_id	movie_name	year_of_release
6	Twister	1997

▼ Click here for Hint

use the command 'insert into table'

▼ Click here for Solution

On the cqlsh run the below command.

```
1 INSERT into movies(
2 movie_id, movie_name, year_of_release)
3 VALUES (6,'Twister',1997);
```

2. Problem:

Modify the release year of Twister to 1996.

▼ Click here for Hint

use the command 'update table set column=value where column=value'

▼ Click here for Solution

On the cqlsh run the below command.

```
1 UPDATE movies
2 SET year_of_release = 1996
3 WHERE movie_id = 6;
```



3. Problem:

Delete the movie Twister.

▼ Click here for Hint

use the command 'delete from ..'

▼ Click here for Solution

On the cqlsh run the below command.

```
1 DELETE from movies
2 WHERE movie_id = 6;
```



4. Problem:

Drop the training keyspace.

▼ Click here for Hint

use the 'drop keyspace' command

▼ Click here for Solution

On the cqlsh run the below command.

```
1 drop keyspace training;
```



5. Problem:

Disconnect from the cassandra server.

▼ Click here for Hint

Refer to the [Using the CQL Shell \(cqlsh\) lab](#).

▼ Click here for Solution

Run the below command on the terminal.

```
1 exit
```

